Experiment No. 02

Aim:- To install and configure MongoDB to execute NoSQL commands.

Theory:-

What is MongoDB?
MongoDB is a NoSQL (Not only SQL) database that stores large volumes of data in the form of documents. MongoDB removes the concept of "rows" of conventional and relational data models by introducing "documents." This offers the developers the flexibility to work with evolving data models.

What is MySQL?
MySQL is a free, open-source, relational database management system that stores data in the form of tables containing rows and columns. It uses RDBMS to ensure referential integrity between the rows of a table and interprets queries to fetch information from the database.

MySQL vs. MongoDB: One-on-one Comparision
Now that you know the objectives of these database management systems, let's look at some of the differences between them.

Feature	MySQL	MongoDB
Data Structure	It stores each individual record as a table cell with rows and columns	It stores unrelated Data in JSON-like documents

Schema	MySQL requires a schema definition for the tables in the database	MongoDB doesn’t any requireprior schema
Languages	Supports Structured Language (SQL) Query	Supports JSON Query Language to work with data
Foreign Key	Supports the usage of Foreign keys	Doesn’t support the of usageForeign keys
Replication	Supports master-slave replication and master-master replication	Supports sharding and replication
Scalability	SQL Database can be scaledvertically	MongoDB database can be scaledboth vertically and horizontally

Join Operation	Supports Join operation	Doesn’t support Join operation
Performance	Optimized for high performance joins across multiple tables	Optimized for write performance
Risks	Prone to SQL injection attack	Since there’s no schema, lesserrisks involved
Community Support	There are currently (always increasing) about 222k repositories and 7Million commits on GitHub for support on MySQL	There are currently (always increasing) about 177k repositories and 923k commits on GitHub for support on MongoDB

Which One to Choose?

Applications, like an accounting system that requires multi-row transactions, would be better suited for a relational database. MySQL is an excellent choice if you have structured data and need a traditional relational database.	MongoDB is well-suited for real-time analytics, content management, the Internet of Things, mobile, and other types of applications. It is an ideal choice if you have unstructured and/or structured data with rapid growth potential.

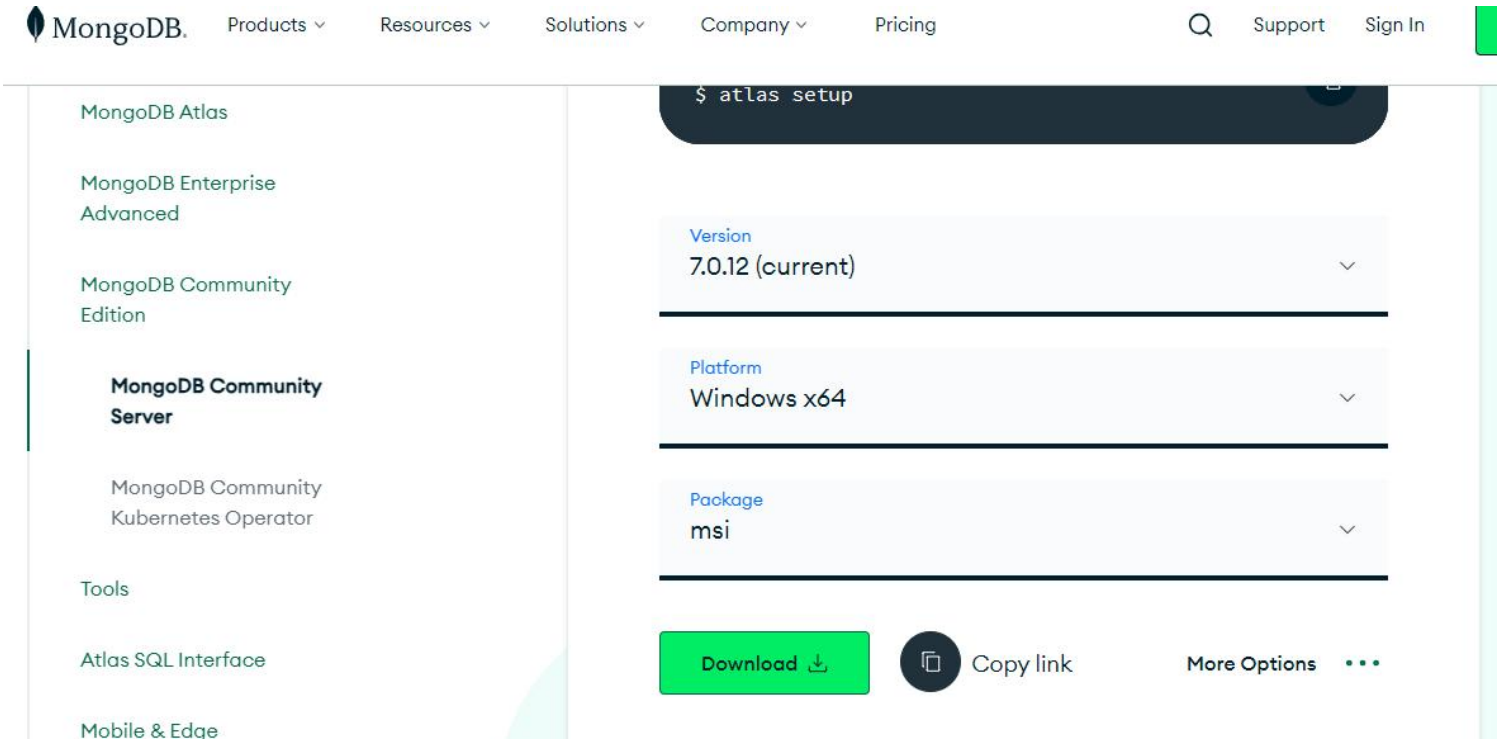
Steps to install and configure MongoDB:-

1. Install MongoDB On Windows

To install MongoDB on Windows, first download the latest release of MongoDB from https://www.mongodb.com/try/download/community. Make sure you get the correct version of MongoDB depending on your Windows version.

1. In the **Version** dropdown, select the version of MongoDB to download.
2. In the **Platform** dropdown, select **Windows**.
3. In the **Package** dropdown, select **msi**.
4. Click **Download**.

2. Also we Install MongoDB Shell. Click on **https://www.mongodb.com/try/download/shell**



3. Run the MongoDB installer.

For example, from the Windows Explorer/File Explorer:

1. Go to the directory where you downloaded the MongoDB installer (.msi file). Bydefault, this is your Downloads directory.

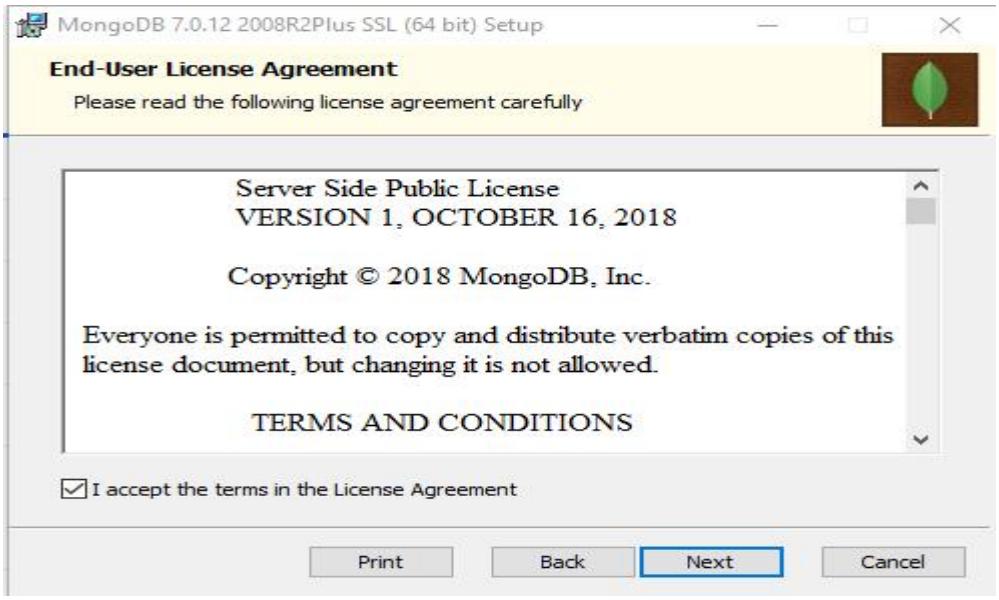
2 Double-click the .msi file.



3 Follow the MongoDB Community Edition installation wizard.

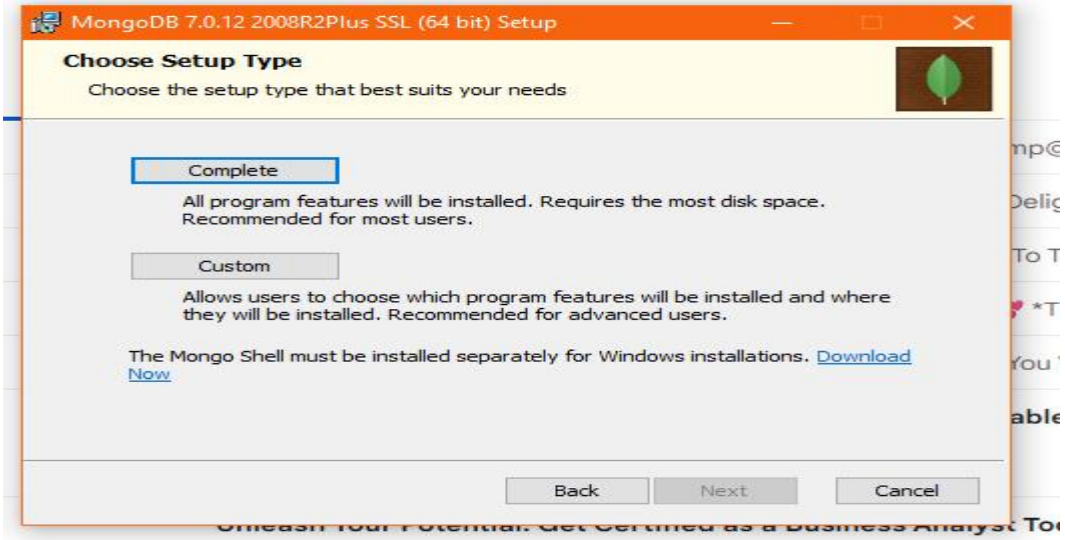
The wizard steps you through the installation of MongoDB and MongoDB Compass.

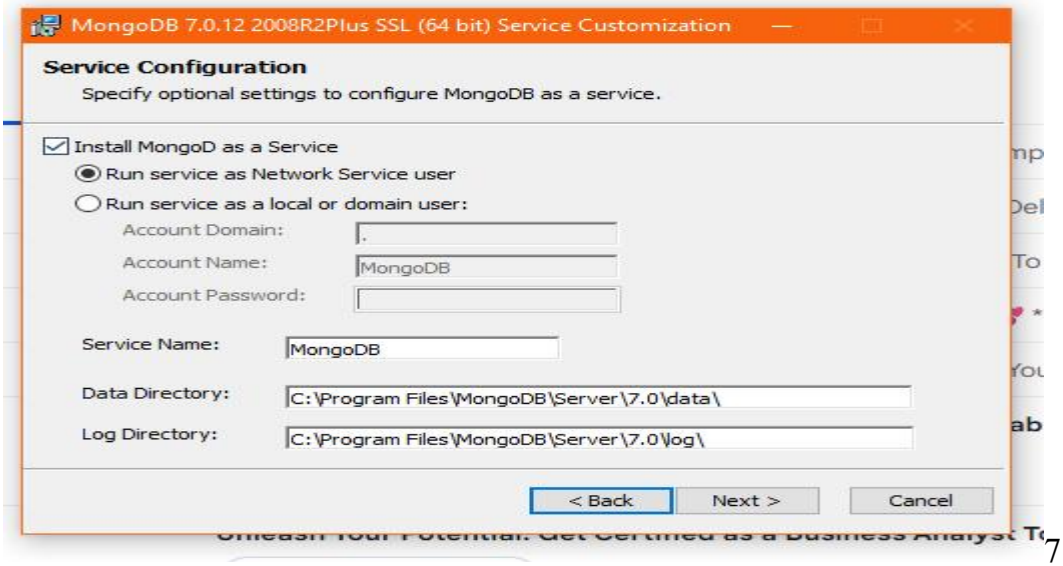
1. **Choose Setup Type**
   You can choose either the **Complete** (recommended for most users) or **Custom** setup type. The **Complete** setup option installs MongoDB and the MongoDB tools to the default location. The **Custom** setup option allows you to specify which executables are installed and where.



4 Service Configuration

Starting in MongoDB 5.0.9, you can set up MongoDB as a Windows service duringthe installation or just install the binaries.

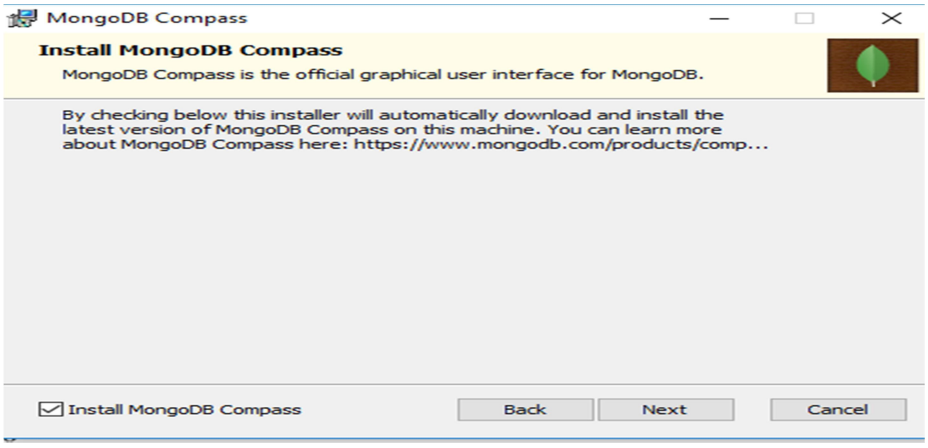




- Select **Install MongoDB as a Service**.
- Select:
  - **Run the service as a Network Service user** (Default)
    This is a Windows user account that is built-in to Windows
- **Service Name**. Specify the service name. The default name is MongoDB. You must choose another name if you already have a service with the specified name.
- **Data Directory**. Specify the data directory, which corresponds to the --dbpath. If the directory does not exist, the installer will create the directory and sets the directory access to the service user.
- **Log Directory**. Specify the Log directory, which corresponds to the --logpath. If the directory does not exist, the installer will create the directory and sets the directory access to the service user.

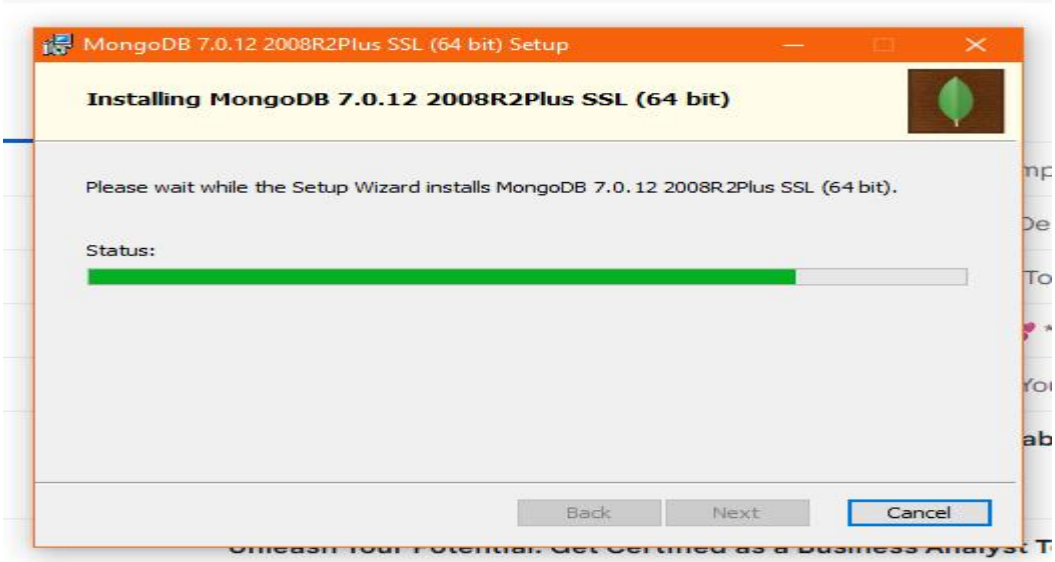
5 Install MongoDB Compass

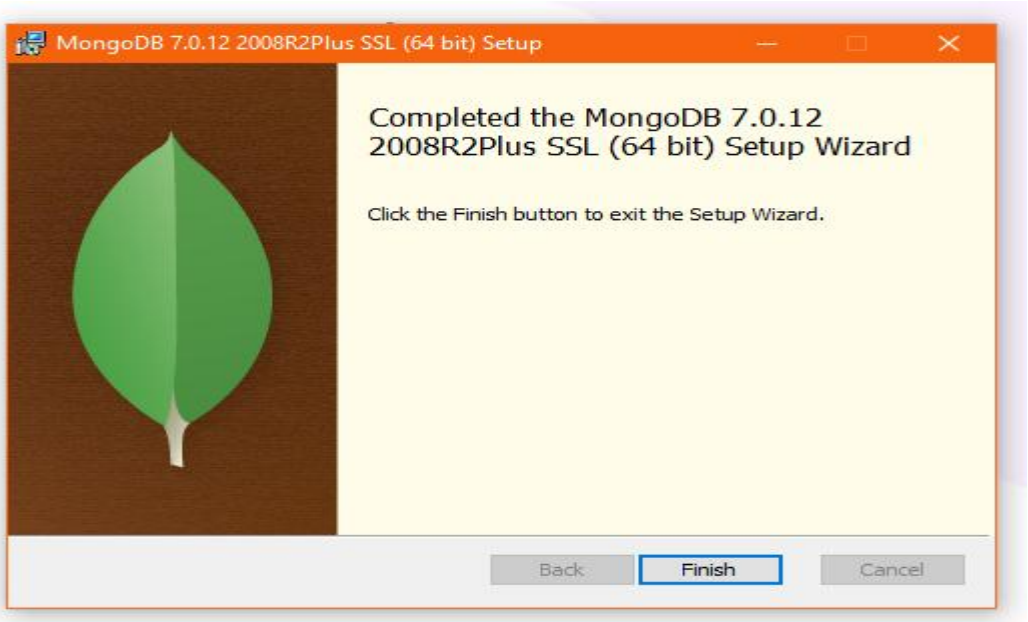
Optional. To have the wizard install MongoDB Compass, select **Install MongoDBCompass** (Default)

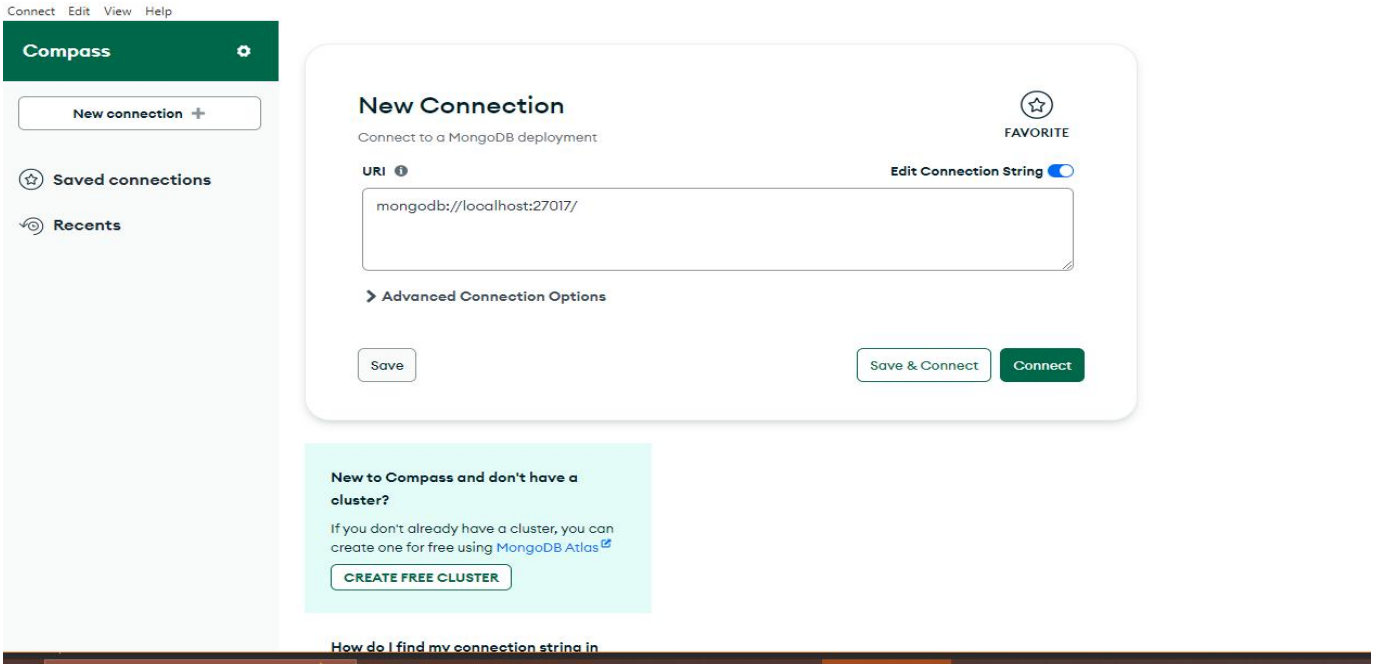




6 When ready, click **Install**.

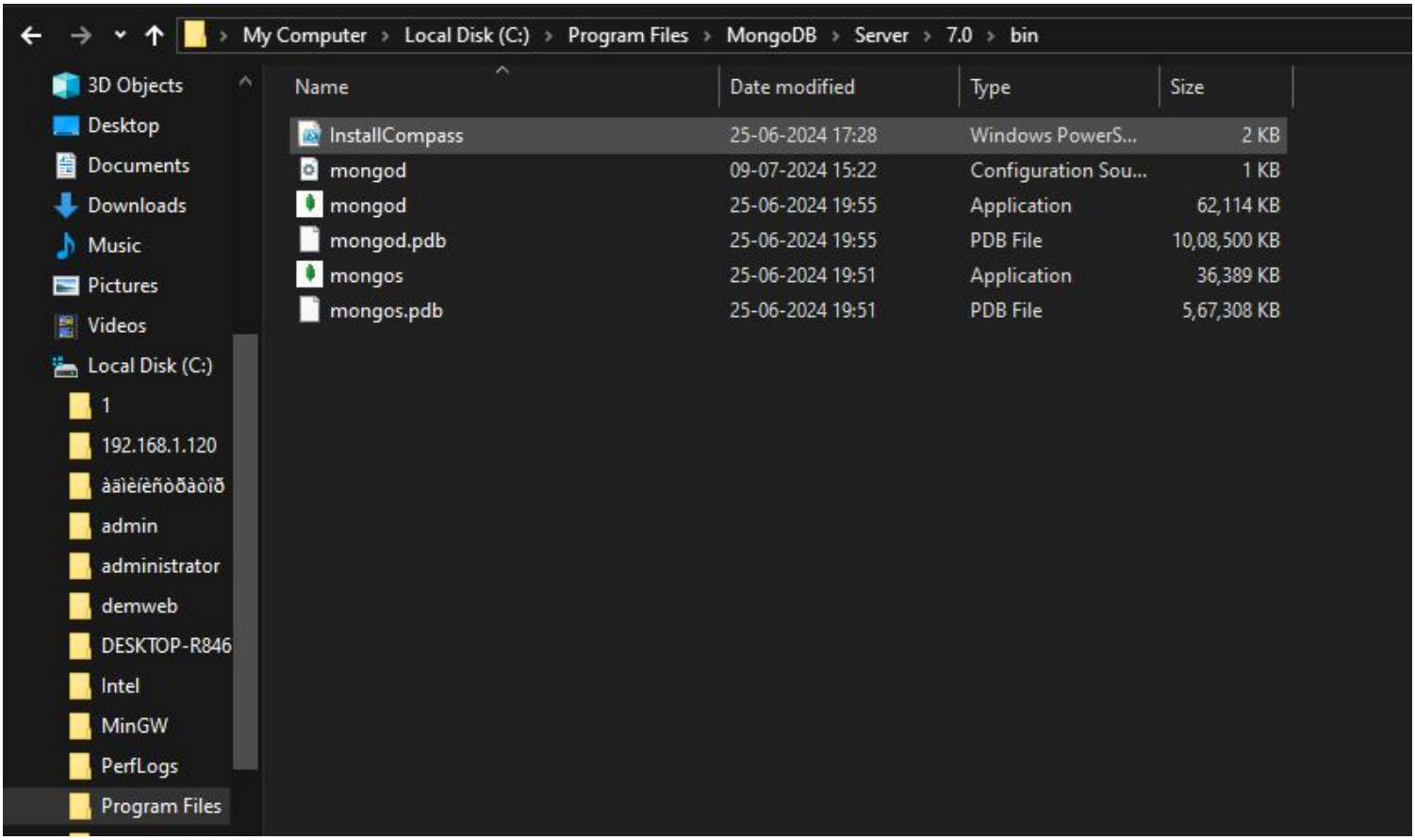



MongoDB 7.0.12 2008R2Plus SSL (64 bit) Setup
Completed the MongoDB 7.0.12 2008R2Plus SSL (64 bit) Setup Wizard
Click the Finish button to exit the Setup Wizard.
Back
Finish
Cancel


Connect Edit View Help
Compass
New connection
Saved connections
Recents
New Connection
Connect to a MongoDB deployment
FAVORITE
URI
Edit Connection String
mongodb://localhost:27017/
Advanced Connection Options
Save
Save & Connect
Connect
New to Compass and don't have a cluster?
If you don't already have a cluster, you can create one for free using MongoDB Atlas
CREATE FREE CLUSTER
How do I find my connection string in

7 Running MongoDB

Go to the location where mongo DB is installed “C:\Program Files\MongoDB\Server\5.0\bin” and open the command prompt at that location and execute the following command in the command prompt.



Commands:

1. **Show All Databases**

Use the below command to get a list of all databases.
show dbs

2. **Create a new database**

To create a new database execute the following command.
use DATABASE_NAME

3. **Know your current selected database**

To know your current working/selected database execute the following command
Db

4. **Create collection**

To create the new collection execute the following commands
db.createCollection(name)

5. **To check collections list**

To get the list of collections created execute the following command
Show collections

6. **Drop collection**

To drop the selected collection execute the following command
>db.COLLECTION_NAME.drop()

7. **Insert document in collection**

>db.COLLECTION_NAME.insert(document)

8. **Get collection document**

To get the list documents in collection execute the following command
>db.COLLECTION_NAME.find()

9. **Update document**

To update the document in collection execute the following command
>db.COLLECTION_NAME.update(SELECTION_CRITERIA, UPDATED_DATA)

10. **Save document**

To save document in collection execute the followingcommand
>db.COLLECTION_NAME.save({_id:ObjectId(),NEW_DATA})

11. **Delete document**

To delete document in selected collection execute the following command
>db.COLLECTION_NAME.remove(DELLETION_CRITTERIA)

Output:-

```
Microsoft Windows [Version 10.0.22621.3880]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>mongod --version
db version v7.0.12
Build Info: {
    "version": "7.0.12",
    "gitVersion": "b6513ce0781db6818e24619e8a461eae90bc94fc",
    "modules": [],
    "allocator": "tcmalloc",
    "environment": {
        "distmod": "windows",
        "distarch": "x86_64",
        "target_arch": "x86_64"
    }
}

C:\Users\Admin>
```

```
Microsoft Windows [Version 10.0.22621.3880]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>mongosh
Current Mongosh Log ID: 66b0b3057dcd3ef452228fb4
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.2.15
Using MongoDB:          7.0.12
Using Mongosh:          2.2.15

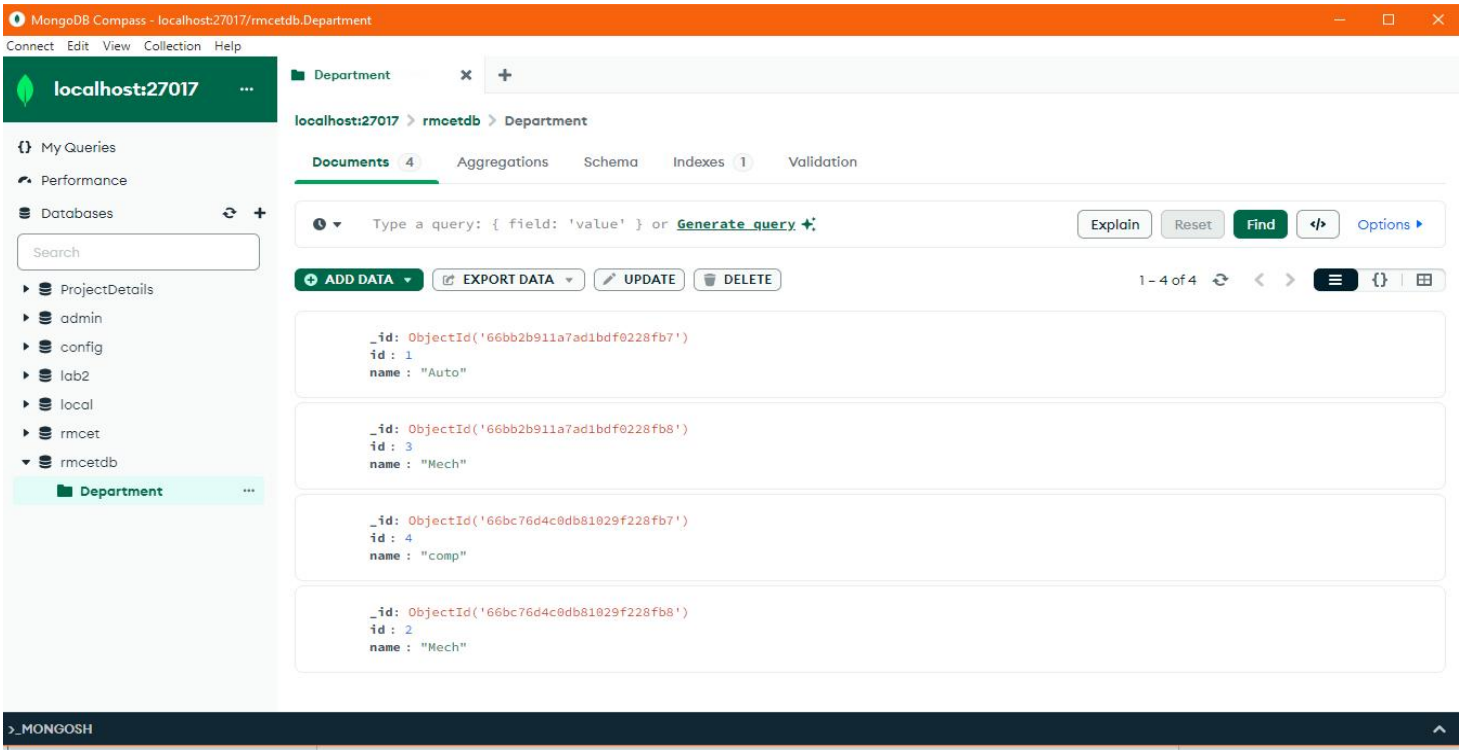
For mongosh info see: https://docs.mongodb.com/mongodb-shell/

------
   The server generated these startup warnings when booting
   2024-08-05T15:47:44.239+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> show dbs
admin        40.00 KiB
config      108.00 KiB
dnyanadadb    8.00 KiB
local        40.00 KiB
test> use rmcetdb
switched to db rmcetdb
rmcetdb> db.createCollection("Student")
{ ok: 1 }
rmcetdb> db.createCollection("Department")
{ ok: 1 }
rmcetdb>  db.Student.insertOne({name:"abc",address:"ambav"})
{
  acknowledged: true,
  insertedId: ObjectId('66b0b3d17dcd3ef452228fb5')
}
rmcetdb>  db.Student.insertOne({name:"pqr",address:"ambav1"})
{
```

```
}
rmcetdb>  db.Department.insertMany([{id:1,name:"comp"},{id:1,name:"Mech"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66b0b4a77dcd3ef452228fb7'),
    '1': ObjectId('66b0b4a77dcd3ef452228fb8')
  }
}
rmcetdb> db.student.findOne({ name: "abc" });
null
rmcetdb> db.Student.findOne({ name: "abc" });
{
  _id: ObjectId('66b0b3d17dcd3ef452228fb5'),
  name: 'abc',
  address: 'ambav'
}
rmcetdb>  db.Department.find().toArray();
[
  { _id: ObjectId('66b0b4a77dcd3ef452228fb7'), id: 1, name: 'comp' },
  { _id: ObjectId('66b0b4a77dcd3ef452228fb8'), id: 1, name: 'Mech' }
]
rmcetdb> show collections
Department
Student
rmcetdb> db.Student.deleteOne({ name: "abc" });
{ acknowledged: true, deletedCount: 1 }
rmcetdb> db.student.findOne({ name: "abc" });
null
```

```
rmcetdb> db.Employee.updateOne(
...    { name: "abc" },          // Query filter to match the document
...    { $set: { address: "new_address" } } // Update operation
...
...
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
rmcetdb> db.Department.updateOne({id:1},{$set:{name:"EXTC"}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
rmcetdb>  db.Department.find().toArray();
[
  { _id: ObjectId('66b0b4a77dcd3ef452228fb7'), id: 1, name: 'EXTC' },
  { _id: ObjectId('66b0b4a77dcd3ef452228fb8'), id: 1, name: 'Mech' }
]
rmcetdb> db.Department.updateOne({name:"Mech"},{$set:{id:2}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
rmcetdb>  db.Department.find().toArray();
[
  { _id: ObjectId('66b0b4a77dcd3ef452228fb7'), id: 1, name: 'EXTC' },
  { _id: ObjectId('66b0b4a77dcd3ef452228fb8'), id: 2, name: 'Mech' }
]
rmcetdb> db.Student.drop();
true
```



Conclusion:-

We installed, configured, and executed NoSQL commands in MongoDB.